

Numerical & mathematical skills for neuroscience

PS5210 2025-26

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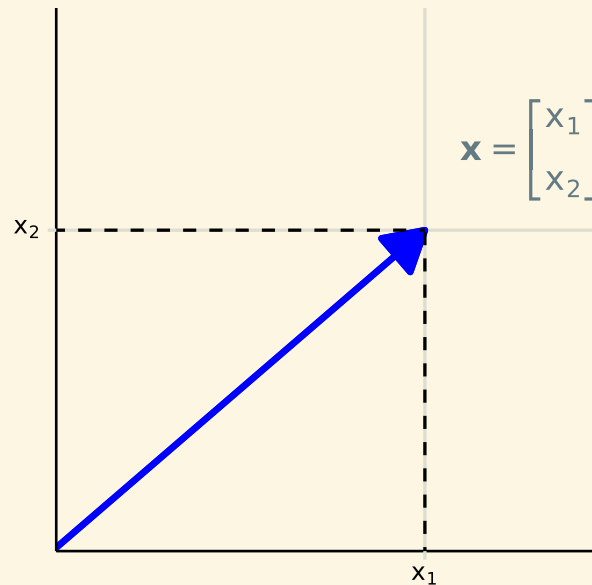
Part 2: introduction to linear algebra

Vocabulary

- vector
- matrix
- linear combination
- linear (in)dependence
- basis
- span
- transformation
- eigenvalues & eigenvectors

Vectors

- A list of n numbers representing a point, or an "arrow", in n -dimensional space
- The vector $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2$ represents a point/arrow in a 2-dimensional space



Vectors

A 3-dimensional
vector:

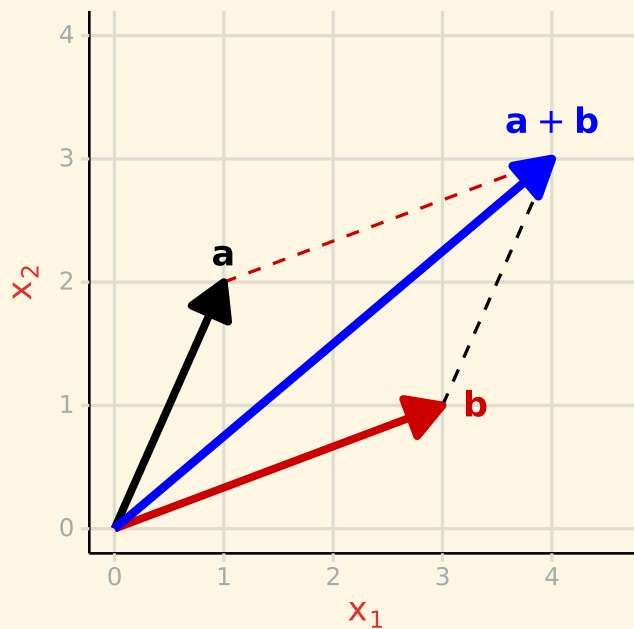
$$\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\vec{x} \in \mathbb{R}^3$$

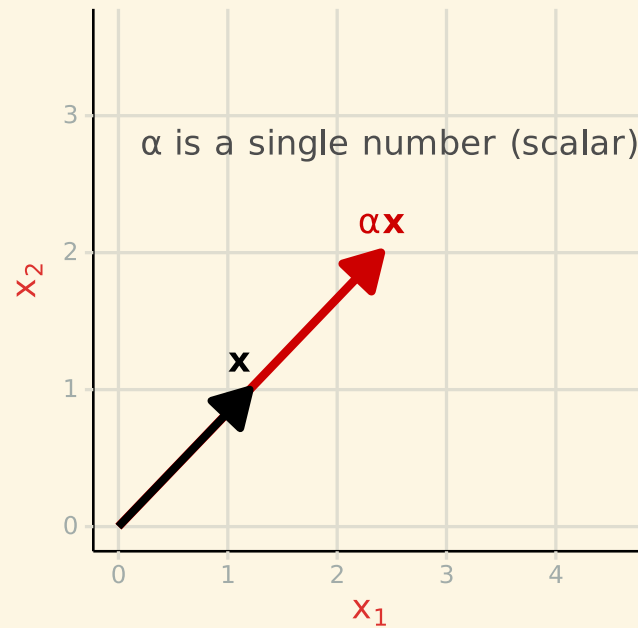
Sum of vectors

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \vec{a + b} = \begin{bmatrix} 1 + 3 \\ 2 + 1 \end{bmatrix}$$

$$\vec{a + b} = \vec{b + a}$$



Multiplication of a vector with a single number (*scalar*)



$$\alpha \vec{x} = \alpha \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_N \end{bmatrix}$$

Elementwise vector multiplication (Hadamard product)

$$\vec{x} \circ \vec{y} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \circ \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_1 y_1 \\ \vdots \\ x_n y_n \end{bmatrix}$$

In Matlab, you can achieve this with `x .* y`

Linear combination of vectors

$$c\vec{v} + d\vec{w} = c \begin{bmatrix} 1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} c + 2d \\ c + 3d \end{bmatrix}$$

*How do you describe the set of **all** possible combinations of $\vec{v}$ and $\vec{w}$?*

This is referred to as the **span** of the set of vectors $\{\vec{v}, \vec{w}\}$.

In this case the combinations of $\vec{v}$ and $\vec{w}$ fill a whole two-dimensional plane.

Matrix-vector multiplication

Multiplying a matrix by a vector is like taking linear combinations of the matrix's columns

$$c\vec{v} + d\vec{w} = c \begin{bmatrix} 1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} c + 2d \\ c + 3d \end{bmatrix}$$

$$\mathbf{A}\vec{x} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} c + 2d \\ c + 3d \end{bmatrix}$$

Linear models

The dependent variable y is modeled as a weighted combination of the predictor variables, plus an additive error ϵ

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

$$\epsilon \sim \mathcal{N}(0, \sigma^2)$$

Linear models: matrix notation

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & \dots & x_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m1} & \dots & x_{mn} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_n \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_m \end{pmatrix}$$

Linear independence

$$\mathbf{A}\vec{x} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} c + 2d \\ c + 3d \end{bmatrix}$$

$$\mathbf{B}\vec{x} = \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} c + 3d \\ c + 3d \end{bmatrix}$$

How does the span of the columns of A differ from the span of the columns of B?

A set of vector is said to be *linearly independent* if no vector in the set that is equal to a linear combination of the others.

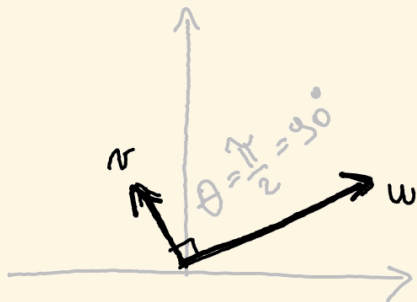
Vector dot product

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2$$

IS A SCALAR

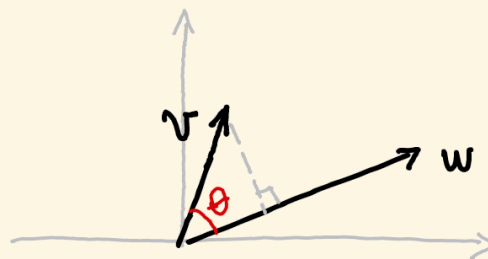
$$v = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \quad w = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$v \cdot w = -4 + 4 = 0$$



$$v = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad w = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$v \cdot w = 5 + 6 = 11$$



$$\cos \theta = \frac{v \cdot w}{\|v\| \|w\|}$$

$$\|v\|^2 = v \cdot v = v_1^2 + v_2^2$$

Vector dot product

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2$$

IS A SCALAR

FOR ANY $\vec{v}, \vec{w}$

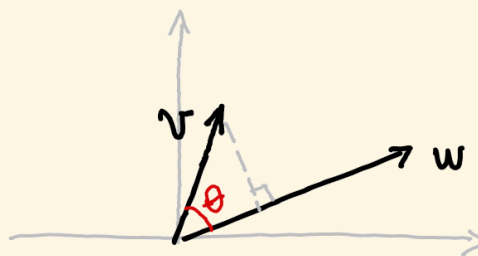
$$v \cdot w = 0 \longrightarrow \theta = \frac{\pi}{2} = 90^\circ$$

$$v \cdot w > 0 \longrightarrow \theta < \frac{\pi}{2}$$

$$v \cdot w < 0 \longrightarrow \theta > \frac{\pi}{2}$$

$$v = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad w = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$v \cdot w = 5 + 6 = 11$$



$$\cos \theta = \frac{v \cdot w}{\|v\| \|w\|}$$

$$\|v\|^2 = v \cdot v = v_1^2 + v_2^2$$

Matrix multiplication

- How do we multiply two matrices?
- While there is only one correct way of doing it, we can do it in 2 slightly different ways:
 1. as a series of dot products;
 2. as a "combination of combinations" of columns.
- This is the default operation for the symbol $\star$ in Matlab.

Matrix multiplication

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix} \quad AB = ?$$

1) MATRIX MULTIPLICATION AS A SERIES OF DOT PRODUCTS

$$(\text{ROWS of } A) \cdot (\text{COLUMNS of } B)$$

$$\begin{aligned} AB &= \begin{bmatrix} \text{row}_1 \cdot \text{col}_1 & \text{row}_1 \cdot \text{col}_2 \\ \text{row}_2 \cdot \text{col}_1 & \text{row}_2 \cdot \text{col}_2 \end{bmatrix} = \\ &= \begin{bmatrix} 2+3 & 2+4 \\ 4-3 & 4-4 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

Matrix multiplication

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 2 \\ 3 & 4 \end{bmatrix} \quad AB = ?$$

2) MATRIX MULTIPLICATION COMBINATIONS OF COLUMNS OF A

$$AB = \left[A \begin{bmatrix} 2 \\ 3 \end{bmatrix} + A \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right]$$

$$= \left[2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 4 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right]$$

$$= \begin{bmatrix} 5 & 6 \\ 1 & 0 \end{bmatrix}$$

Matrix multiplication

RULES MATRIX MULTIPLICATION

$$A_{m \times n} B_{n \times p} = C_{m \times p}$$

N. ROWS N. COLUMNS

$$(AB)C = A(BC)$$

ASSOCIATIVITY

$$AB \neq BA$$

NON-COMMUTATIVITY

(except special cases)

Matrix multiplication in neural networks

$$\mathbf{X}_{1 \times n} \mathbf{W}_{n \times m} = \mathbf{Y}_{1 \times m}$$

$$(x_1, \dots, x_n) \begin{pmatrix} w_{1,1} & \cdots & w_{1,m} \\ \vdots & \ddots & \vdots \\ w_{n,1} & \cdots & w_{n,m} \end{pmatrix} = (y_1, \dots, y_m)$$

$$\left[\underbrace{(x_1, \dots, x_n) \begin{pmatrix} w_{1,1} \\ \vdots \\ w_{n,1} \end{pmatrix}}_{\text{the dot product between } \mathbf{X}_{1 \times n} \text{ and 1st column of } \mathbf{W}_{n \times m} \text{ gives } y_1}, \dots, (x_1, \dots, x_n) \begin{pmatrix} w_{1,m} \\ \vdots \\ w_{n,m} \end{pmatrix} \right] = (y_1, \dots, y_m)$$

the dot product between $\mathbf{X}_{1 \times n}$
and 1st column of $\mathbf{W}_{n \times m}$
gives y_1

Matrices as *linear transformations* of vectors

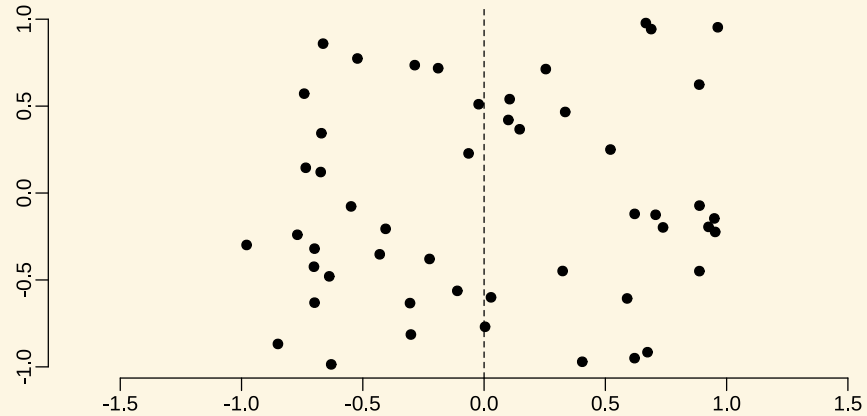
We can think of matrices as functions, which takes vectors as input and apply *transformations* to them.

$$\vec{y} = f(\vec{x}) = A\vec{x}$$

Example: 'stretch' transformation

$$S = \begin{bmatrix} 1 + \delta & 0 \\ 0 & 1 \end{bmatrix}$$

$$S \begin{bmatrix} x_1 & x_2 & \dots \\ y_1 & y_2 & \dots \end{bmatrix} = ?$$

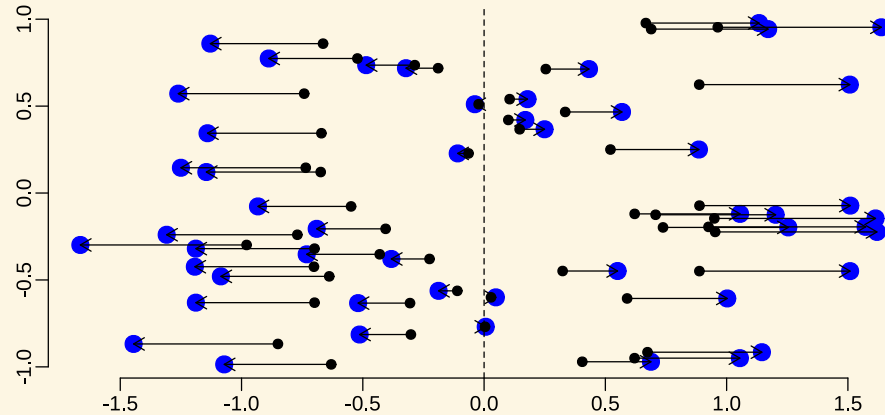


Example: 'stretch' transformation

$$S = \begin{bmatrix} 1 + \delta & 0 \\ 0 & 1 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 + \delta & 0 \\ 0 & 1 \end{bmatrix}$$

$$S\vec{x} = \begin{bmatrix} 1 + \delta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ = \begin{bmatrix} (1 + \delta)x_1 \\ x_2 \end{bmatrix}$$



Special matrices

Identity matrix: $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

For any other matrix A we have that $AI = A = IA$

Special matrices

Inverse matrix: $A^{-1}A = AA^{-1} = I$

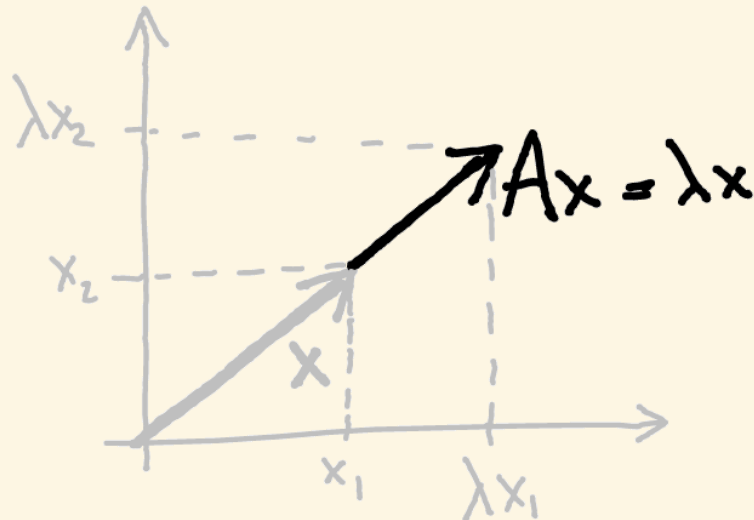
$$S = \begin{bmatrix} 1 + \delta & 0 \\ 0 & 1 \end{bmatrix} \text{ has inverse } S^{-1} = \begin{bmatrix} \frac{1}{1+\delta} & 0 \\ 0 & 1 \end{bmatrix}$$

$$SS^{-1} = \begin{bmatrix} 1 + \delta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{1+\delta} & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1+\delta}{1+\delta} & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Eigenvalues & Eigenvectors

$$Ax = \lambda x$$

λ is the EIGEN VALUE
 x is the EIGEN VECTOR



Eigenvalues & Eigenvectors

$$Ax = \lambda x$$

EIGEN VECTOR
EIGEN VALUE

$$S = \begin{bmatrix} 1+\delta & 0 \\ 0 & 1 \end{bmatrix} \text{ HAS } v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ EIGEN VECTORS}$$

$$S v_1 = \begin{bmatrix} 1+\delta \\ 0 \end{bmatrix} \Rightarrow \lambda_1 = 1+\delta$$

$$S v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \lambda_2 = 1$$

Principal component analysis (PCA)

- A *dimensionality-reduction* method
- Given n -dimensional vectors (e.g. recording from n neurons) as data, we obtain a lower-dimensional representation, by projecting the data on a set of *principal components*, while preserving as much of the data's variation as possible.

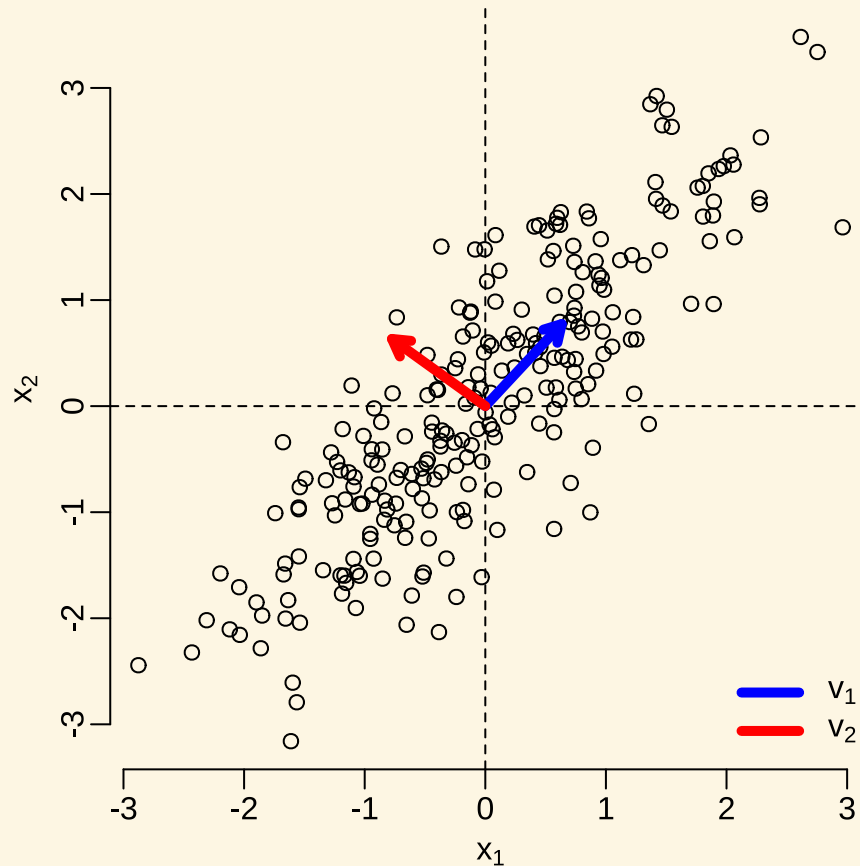
- Variance-covariance matrix:

$$\Sigma = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) & \cdots & \text{Cov}(x_1, x_n) \\ \text{Cov}(x_2, x_1) & \text{Var}(x_2) & \cdots & \text{Cov}(x_2, x_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(x_n, x_1) & \text{Cov}(x_n, x_2) & \cdots & \text{Var}(x_n) \end{bmatrix}$$

- **The eigenvectors of the variance-covariance are the *principal components*.**
- The eigenvector with the largest eigenvalue represents the direction of maximal variation in the data.

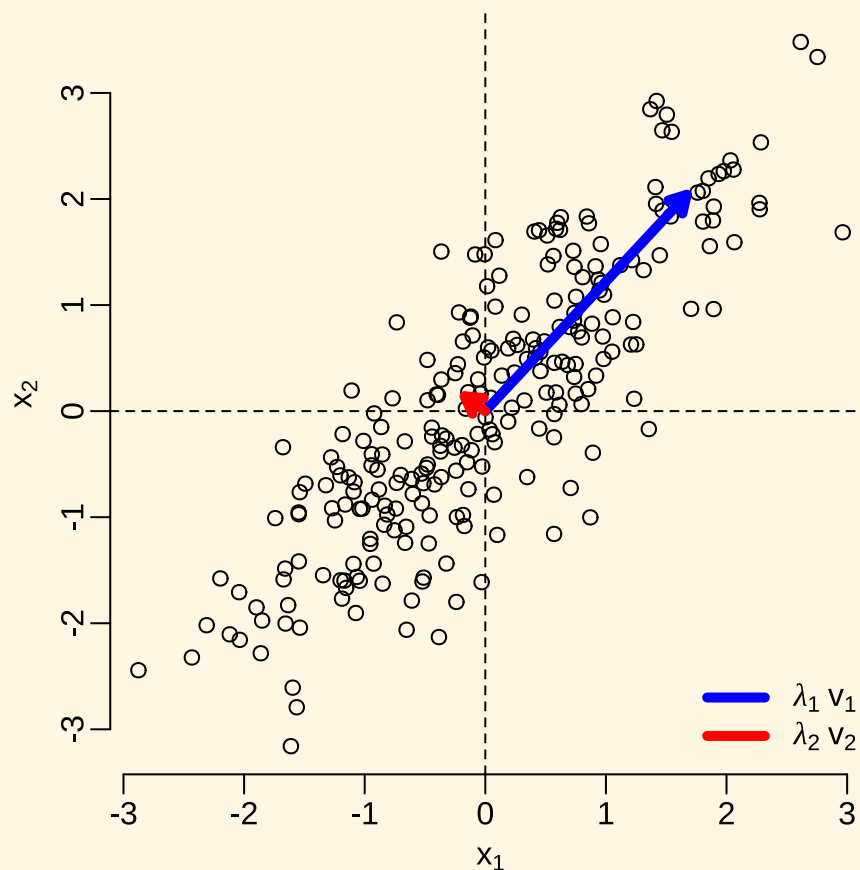
Principal component analysis (PCA)

$$\Sigma = \begin{bmatrix} 1 & 0.99 \\ 0.99 & 1.5 \end{bmatrix}$$



Principal component analysis (PCA)

Now showing the eigenvectors scaled by their respective eigenvalues.



Matlab code for PCA on [Github](#)

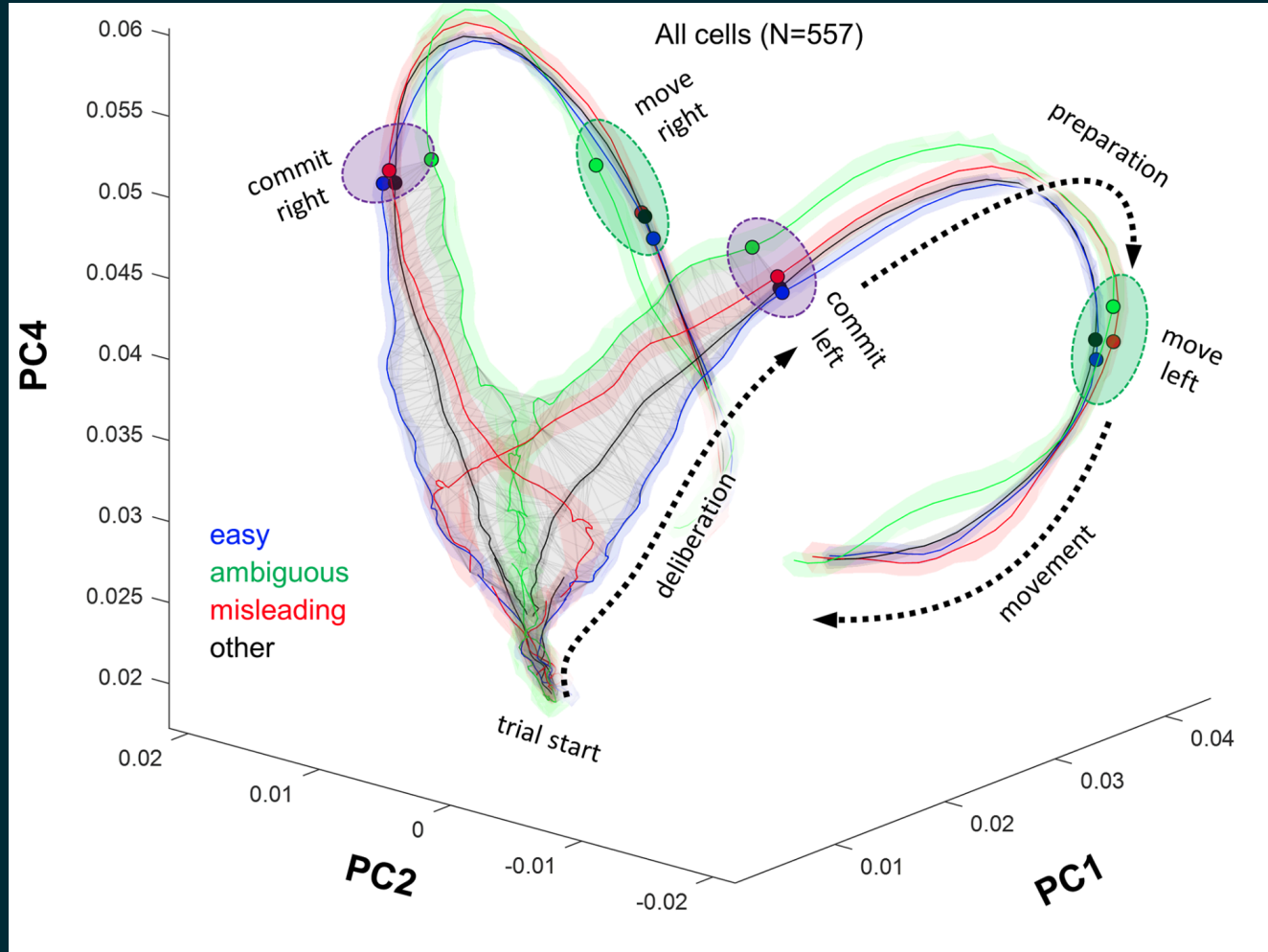
```
% calculate eigenvectors and eigenvalues
[v, lambda] = eig(Sigma);

% sort eigenvectors according to eigenvalues
[lambda, ord] = sort(diag(lambda));
lambda = flip(lambda); ord=flip(ord);
v = v(:,ord);

% v(:,1) is the first principal component
% accounting for the largest fraction of variance in the data
v(:,1)

% plot the first principal component, scaled by eigenvalue
arrow([0,0],v(:,1)*lambda(1), 'linewidth',2);
```

Neural trajectories revealed by PCA



Thura, D., Cabana, J.-F., Feghaly, A., & Cisek, P. (2022). Integrated neural dynamics of sensorimotor decisions and actions. *PLOS Biology*, 20(12), e3001861. <https://doi.org/10.1371/journal.pbio.3001861>

Part 2 summary

Vectors represent points/directions in space; **matrices** can be seen as functions that transform them.

Linear combinations of vectors can create new vectors from old ones; **span** is the space of possible combinations; **linear (in)dependence** tells us when vector carry redundant information.

Matrix-vector multiplication is a linear combination of the matrix columns; **dot products** connect algebra and geometry (gives angular distances between vector orientations); and **matrix multiplication** can be seen as "combination of combinations" of columns or a sequence of dot products.

Eigenvectors are vectors which are fixed in direction under a given linear transformation (a matrix); the scaling factor of these eigenvectors is then called the **eigenvalue**. (Roughly the eigenvalue is a measure of the distortion induced by the transformation and the eigenvectors tell you about how the distortion is oriented.)

PCA uses eigen-decomposition to find directions of maximal variance for constructing a simpler, interpretable representations.

